

Course Content and Resources

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1 The Five Learning Objective Bundles

1.1 The math of QM

- apply an operator to one of its eigenstates
- apply Eulers formula to convert complex exponentials to trig
- apply orthonormality conditions to inner products of basis states
- choose the overall phase of a state to be real and positive
- complex conjugate and transpose to make a ket a bra
- determine the eigensystem of a Hamiltonian
- express the eigenstates of non-z spin measurement in the z basis
- normalize a quantum state
- take an inner product in Dirac or Matrix Notation (complex conjugate and transpose)
- use the unit circle or similar to identify behavior of half-integer multiples of sines and cosines

1.2 Measurement

- correctly identify the possible results of a measurement
- know how measurement affects a superposition state
- know how to calculate the expectation value of an operator
- know how to calculate the probability of a measurement
- know how to calculate the uncertainty of an operator

1.3 Time evolution

- express an initial state in terms of the eigenstates of the Hamiltonians to time evolve
- correctly time-evolve a state using the Hamiltonian
- calculate a time-dependent expectation value
- calculate a time-dependent uncertainty

1.4 Applications

- define the S^2 operator for spin s systems
- define the S_z operator for spin s systems
- identify the Hamiltonian for a particle in an infinite square well
- identify the Hamiltonian for a free particle
- identify the Hamiltonian for an electron in a Hydrogen atom
- identify the Hamiltonian for a particle in a generic potential function
- identify the Hamiltonian for a spin- s particle in a magnetic field

1.5 The wave function

- normalize a state in wave function notation
- find an overlap or probability using wavefunction notation
- calculate an expectation value in wave function notation
- calculate an uncertainty in wave function notation
- solve the energy eigenvalue equation in position representation
- identify and apply the boundary conditions for a particle in a potential

2 Math and Physics Resources

Videos: [Khan Academy's unit on linear algebra](#)

Videos: [Khan Academy's unit on differential equations](#)

Videos: [Khan Academy's unit on multivariable calc](#)

Videos: [Khan Academy's unit on probabilities](#)